

CHAPTER 3(WI)

Business Cycle Measurement

Roadmap



1. Introduction to Business Cycle Measurement
2. Nominal Variables
3. Labor Market Variables.

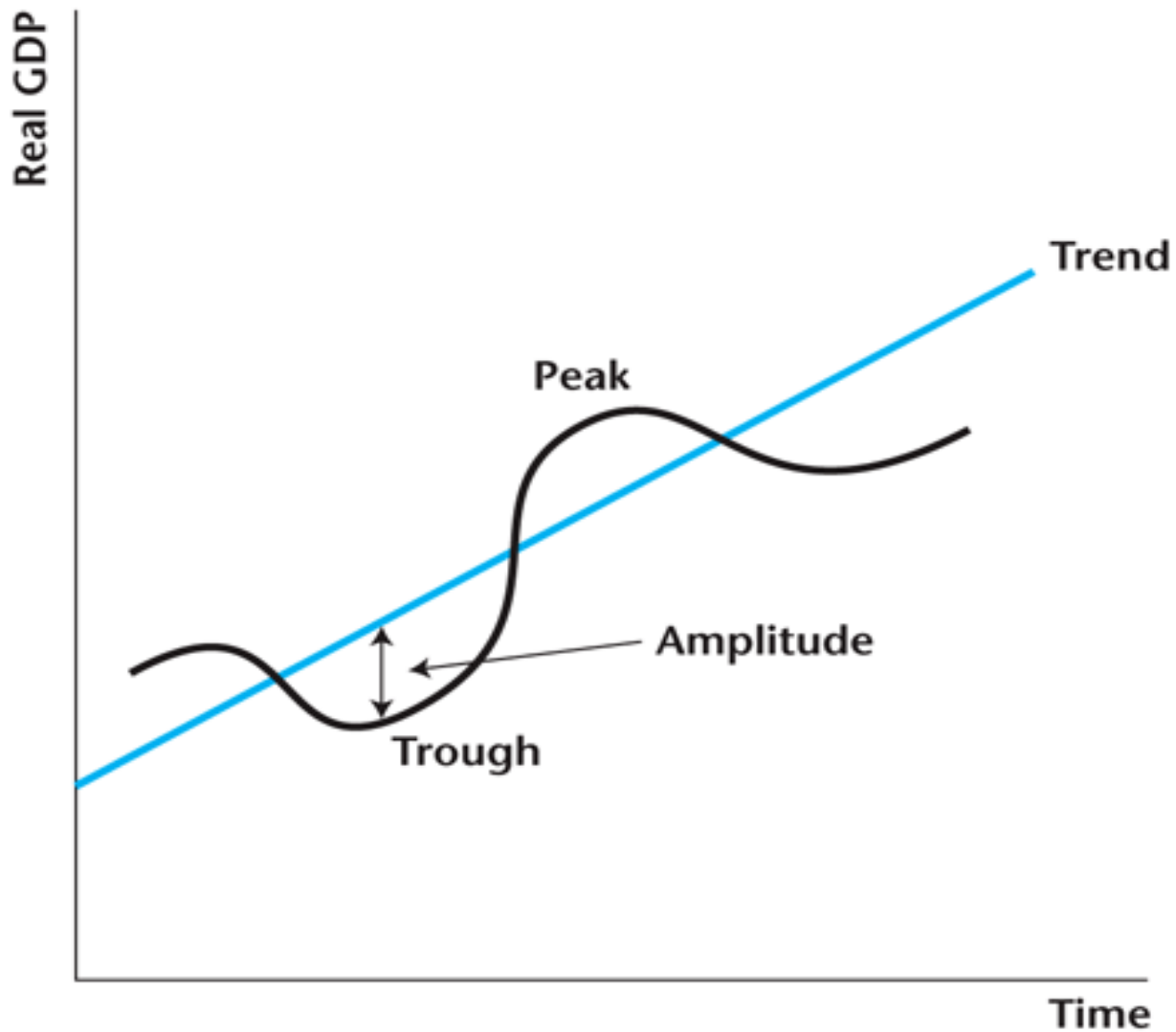
Topics in Business Cycle Measurement

- What are Business Cycles?
- Irregularities in GDP fluctuations
- Regularities of co-movement
- Behavior of Key Macroeconomic Variables.
 - Nominal Variables
 - Labor Market Variables
 - Seasonal Adjustment
- Summary

Business Cycles

- Business Cycles are *fluctuations about trend in real GDP*.
- The turning points in the deviations of real GDP from trend are *peaks and troughs*.
- Persistent positive deviations from trend are *booms* and persistent negative deviations from trend are *recessions*.
- *Amplitude*: difference between peak and trend.
- *Length*: time between one peak and the next.

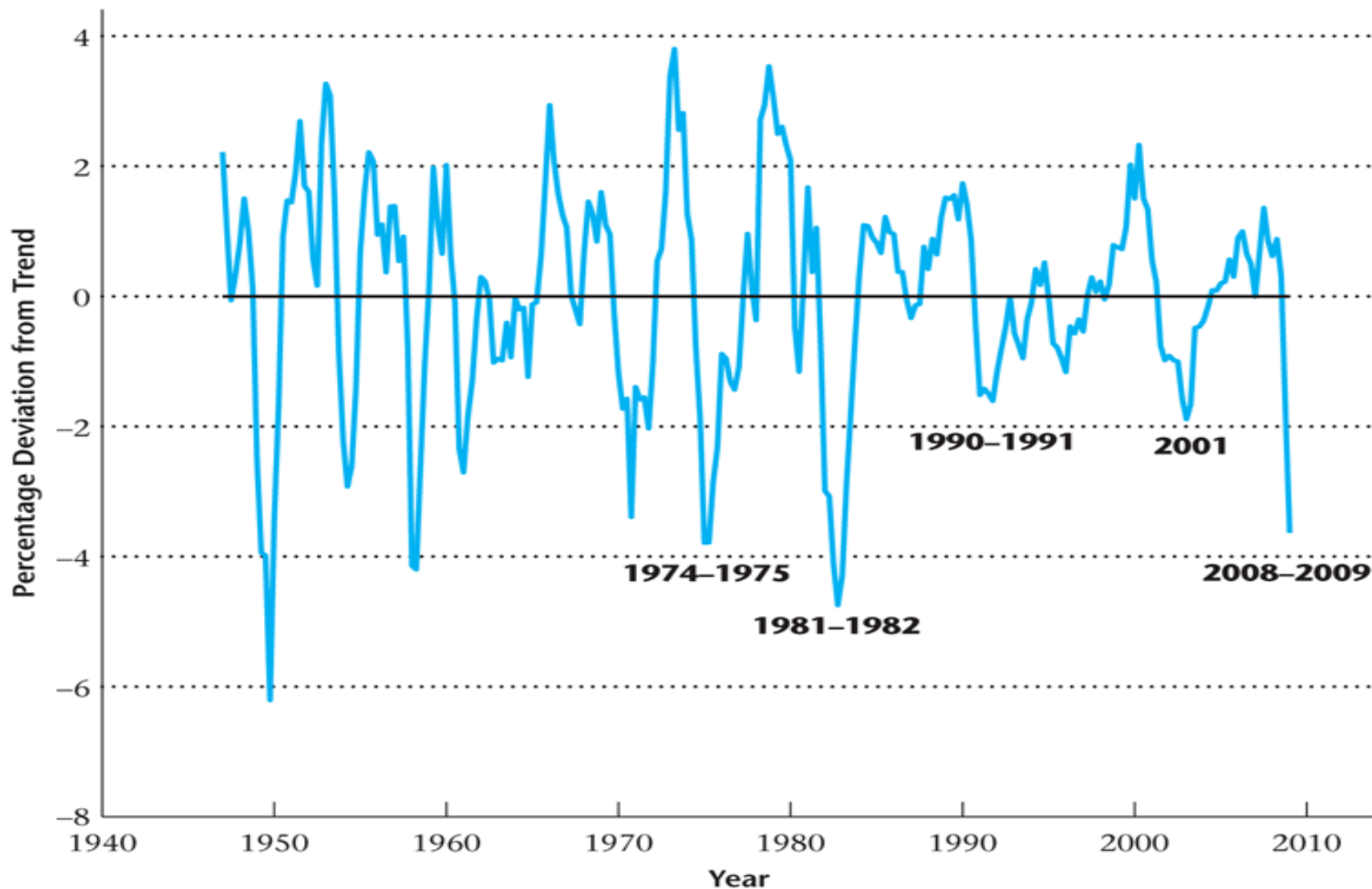
Figure 3.1: Idealized Business Cycles



Irregularities of Deviations From Trend in Real GDP

- The fluctuations in GDP about trend are quite choppy.
- There is *no regularity* in the amplitude of fluctuations in real GDP about trend.
- There is *no regularity* in the frequency of fluctuations in real GDP about trend.

Figure 3.2: Percentage Deviations from Trend in Real GDP from 1947–2009



Regularities

- Regularities imply relationships between macroeconomic variables across business cycles.
- *Variation/Volatility*: standard deviation.
- *Persistence*: autocorrelation (between times).
- *Correlation* between variables.
- *Leads and lags*.

Figure 3.3: Time Series Plots of x and y

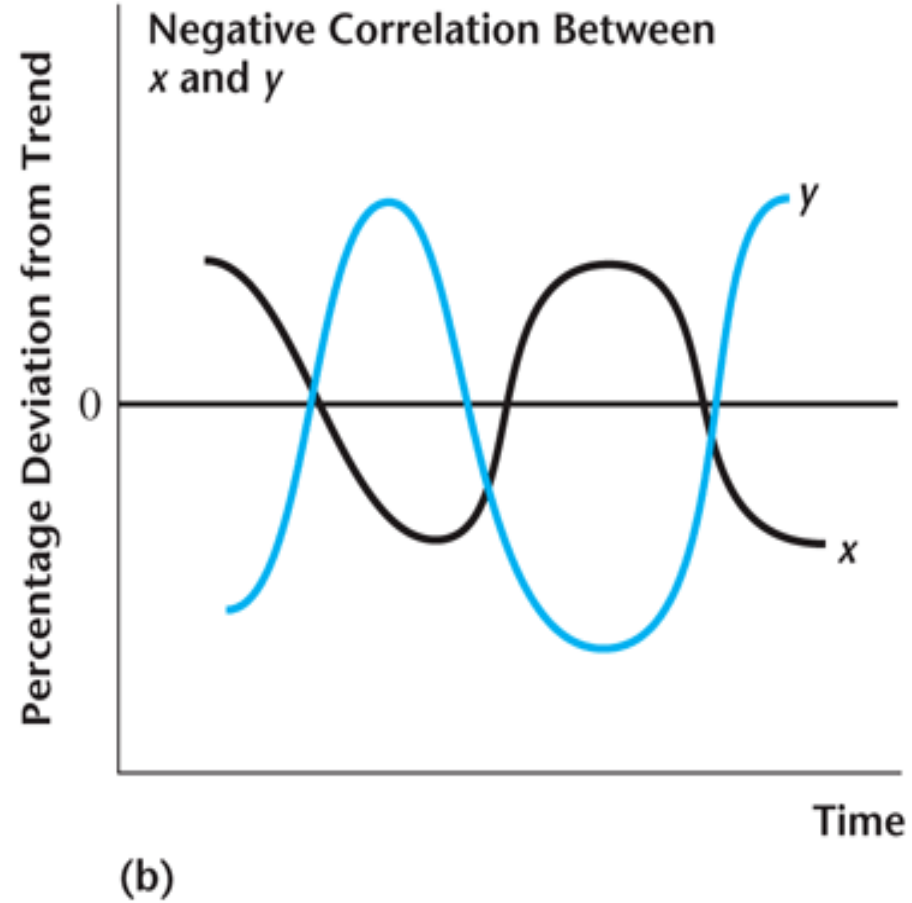
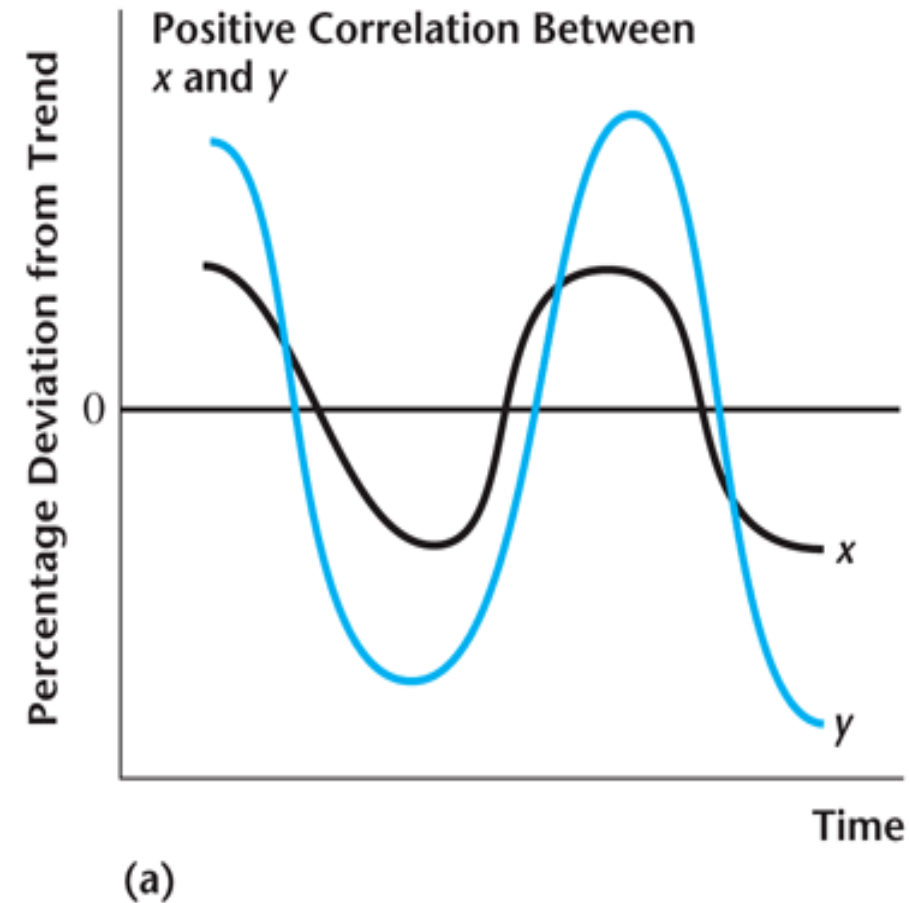
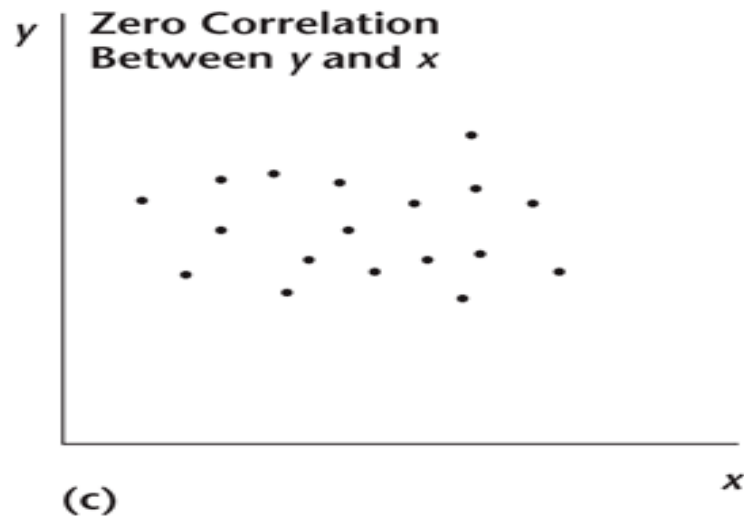
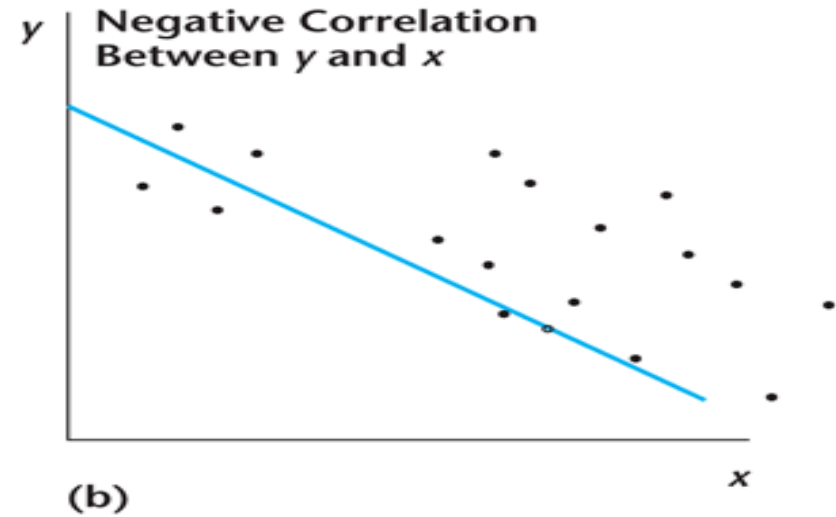
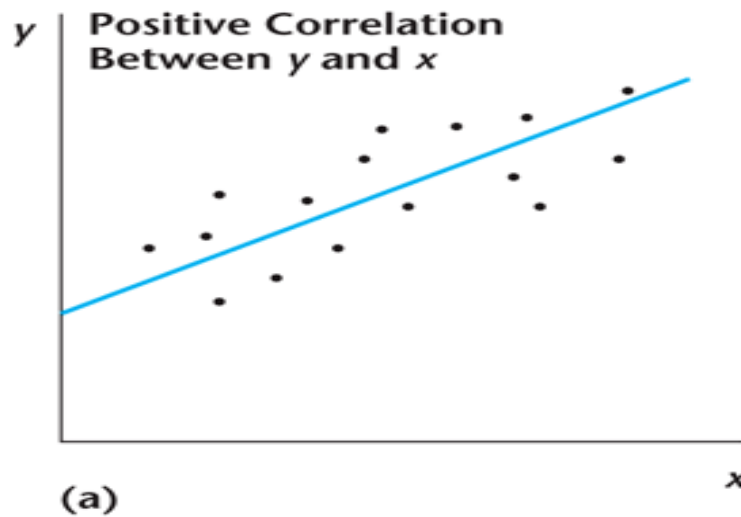


Figure 3.4: Correlations Between Variables y and x



Correlation with Real GDP



- If the deviations from trend in a macroeconomic variable are positively (negatively) correlated with the deviations from trend in real GDP, then that variable is *procyclical* (*countercyclical*).
- If a macroeconomic variable is neither procyclical nor countercyclical, it is *acyclical*.

Figure 3.5: Imports and GDP

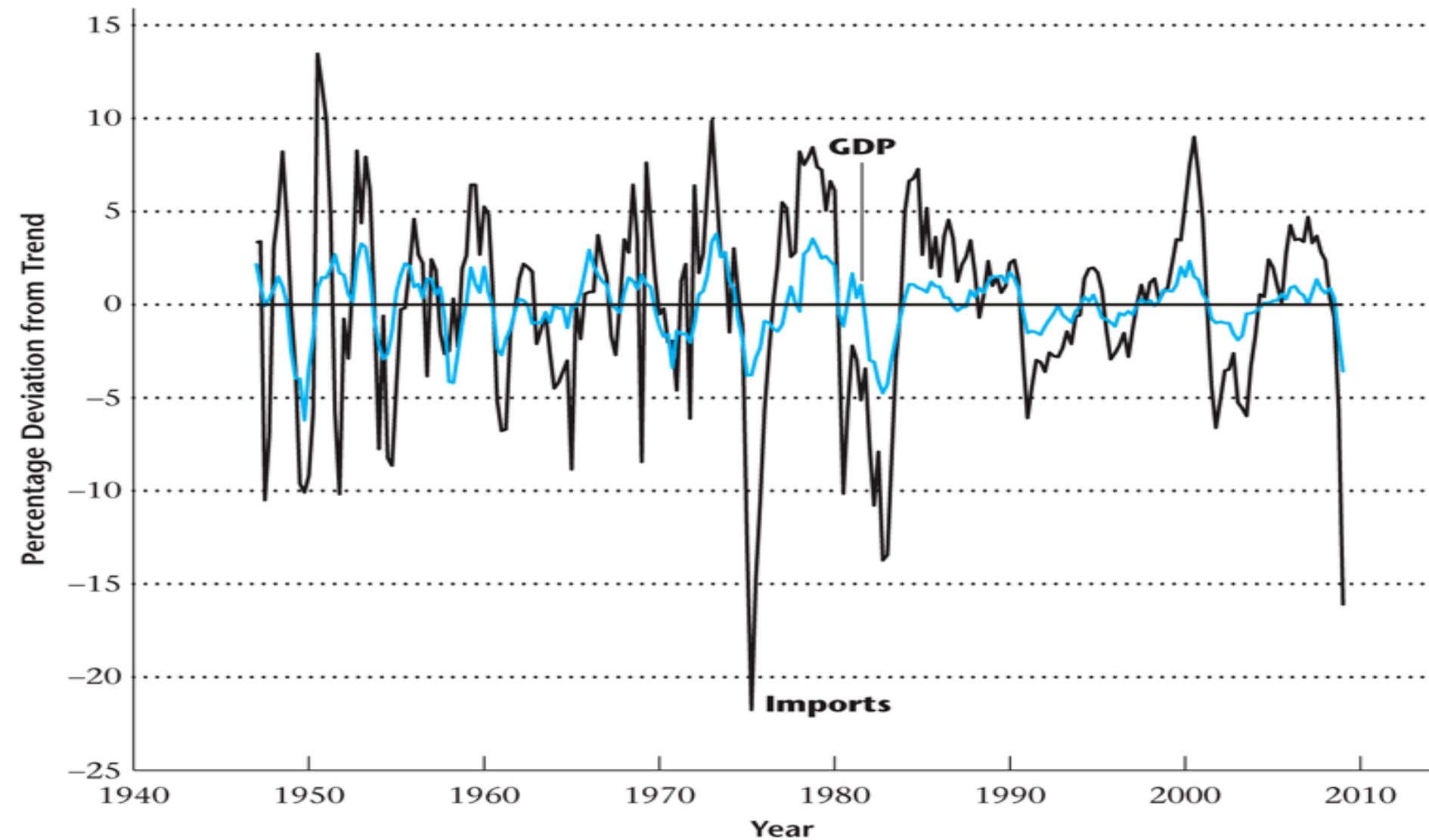


Figure 3.6: Scatter Plot of Imports and GDP

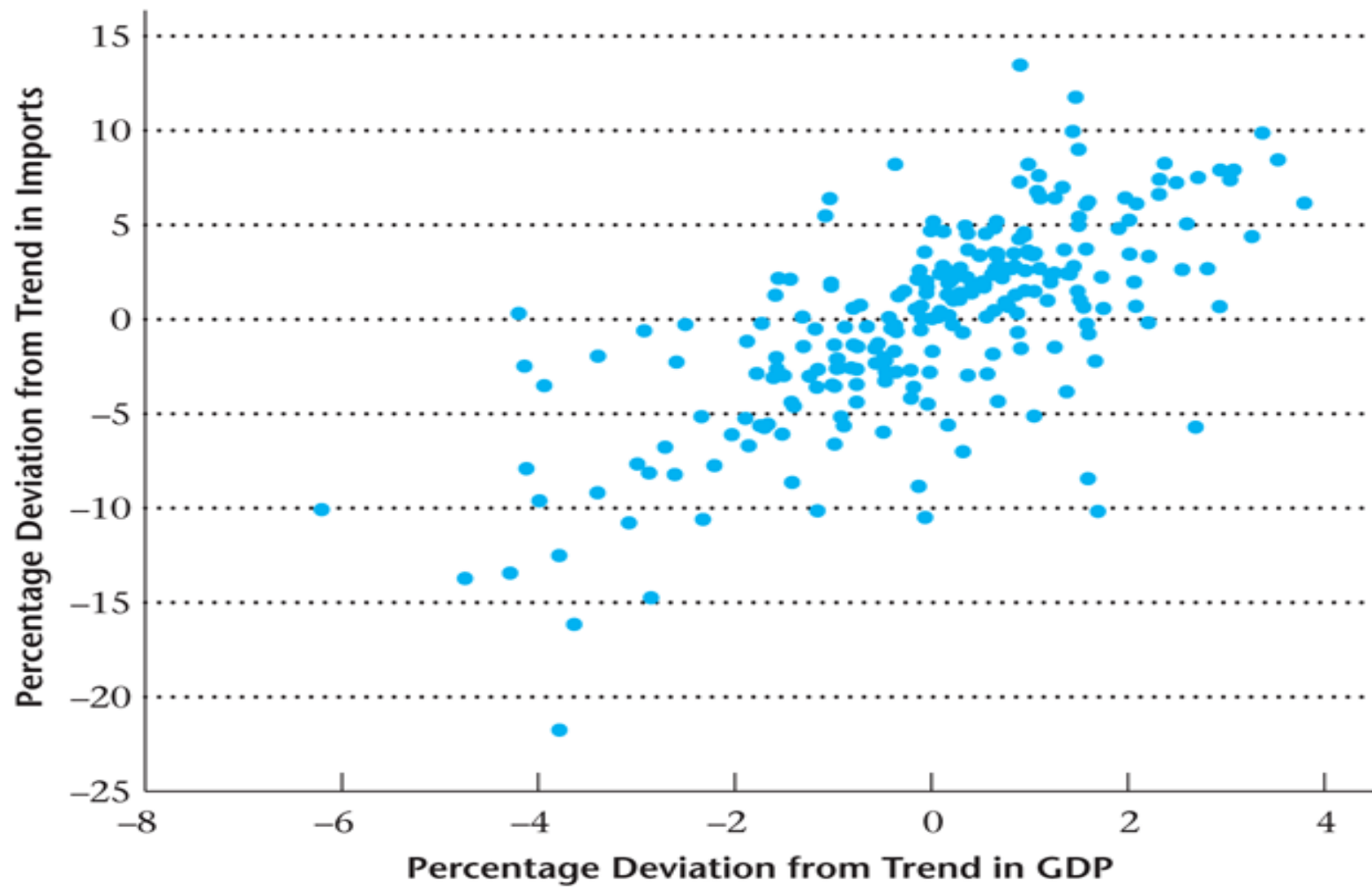


Figure 3.7: Leading and Lagging Variables

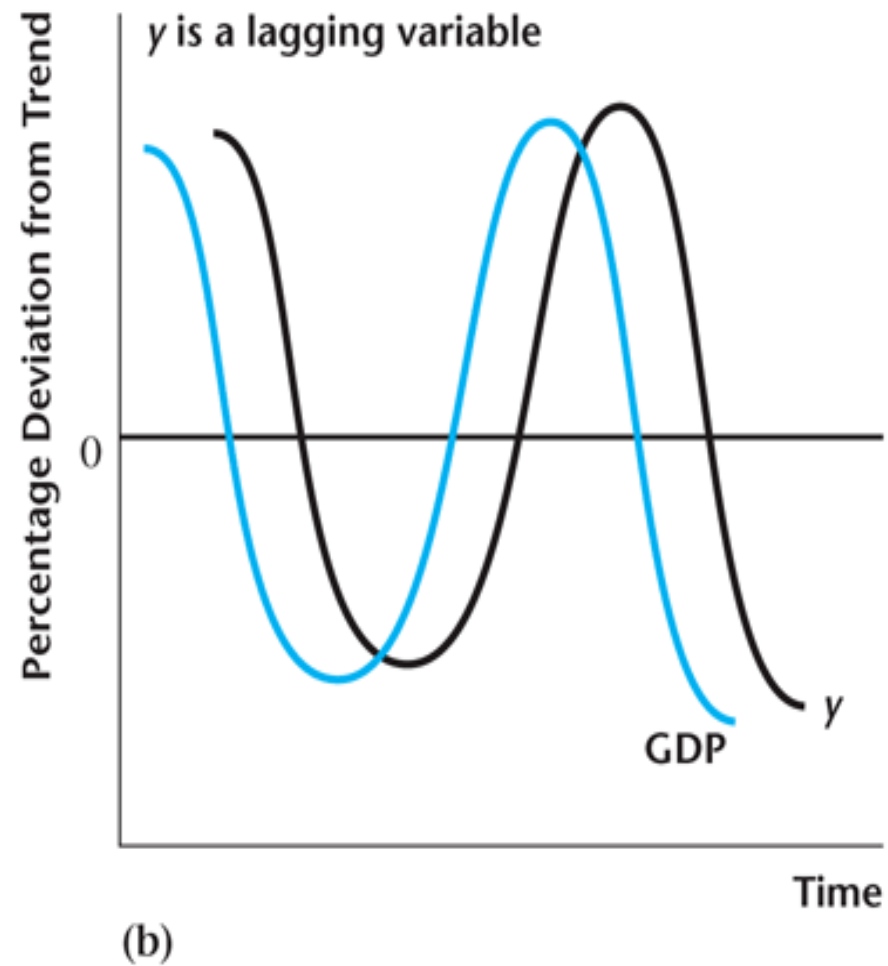
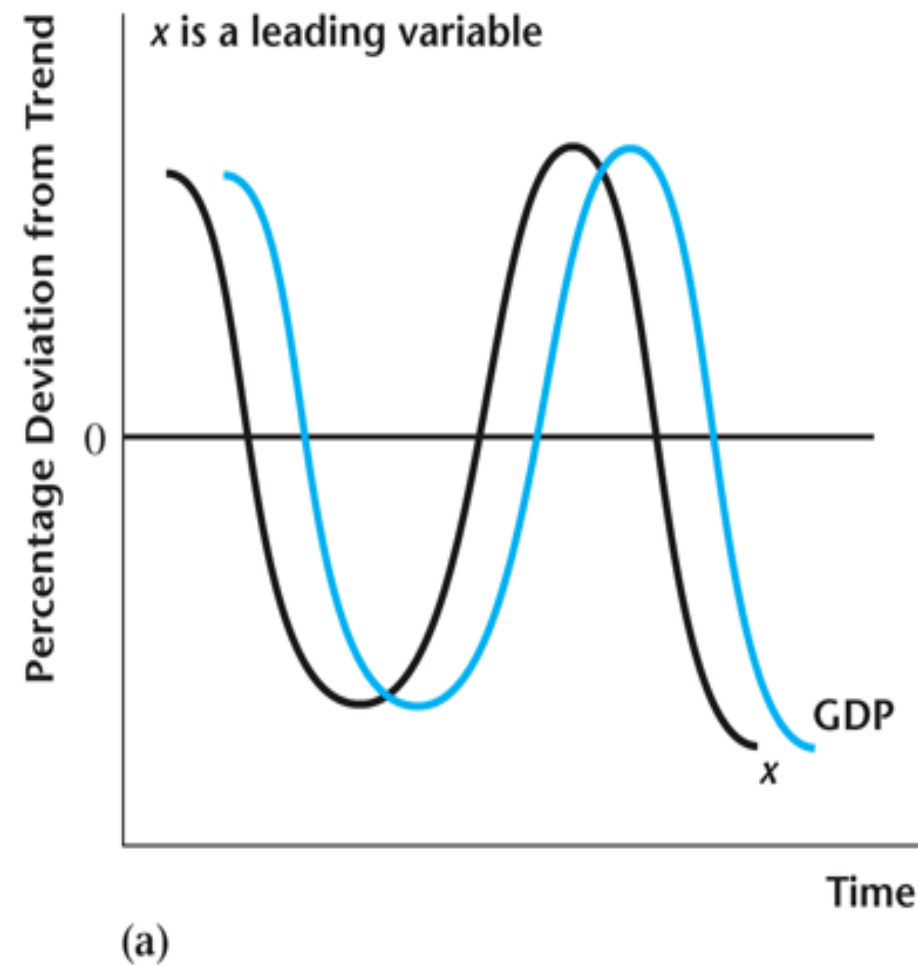
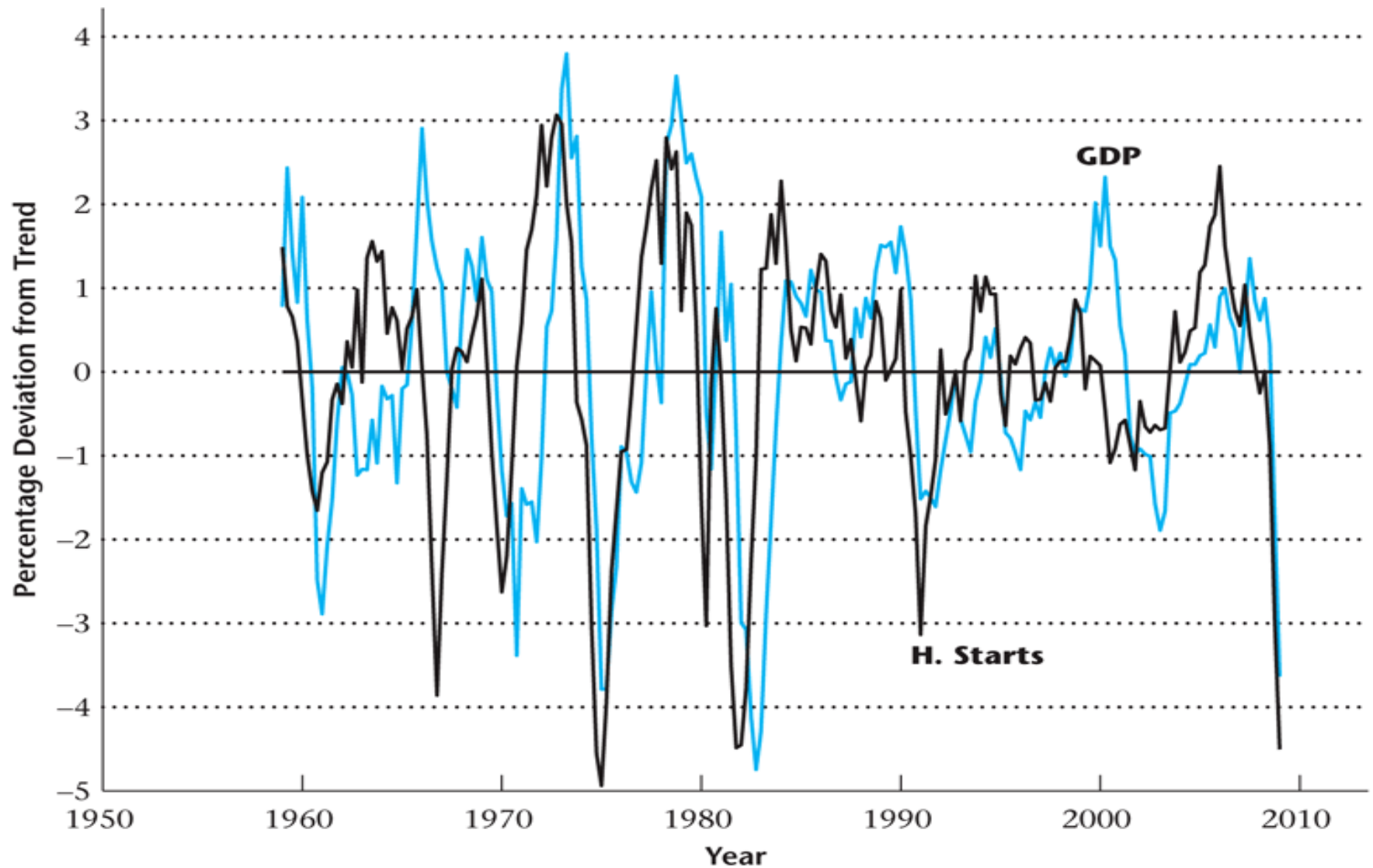


Figure 3.8: Percentage Deviations in Real GDP (colored line) and Housing Starts (black line), for 1959–2009



Behavior of Key Macroeconomic Variables

- GDP components($Y=C+I+G+NX$): consumption, investment, government spending, exports, and imports.
- Nominal variables: price level(CPI, GDP deflator), money supply, interest rates, asset prices.
- Labor market variables: employment, unemployment, wages, labor productivity.

Table 3.2: Summary of Business Cycle Facts

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	Cyclical	Lead/Lag	Variation Relative to GDP
Consumption	Procyclical	Coincident	Smaller
Investment	Procyclical	Coincident	Larger
Price Level	Countercyclical	Coincident	Smaller
Money Supply	Procyclical	Leading	Smaller
Employment	Procyclical	Lagging	Smaller
Real Wage	Procyclical	?	?
Average Labor Productivity	Procyclical	Coincident	Smaller

Figure 3.9: Percentage Deviations from Trend in real Consumption (black line) & GDP (colored) 1947–2009

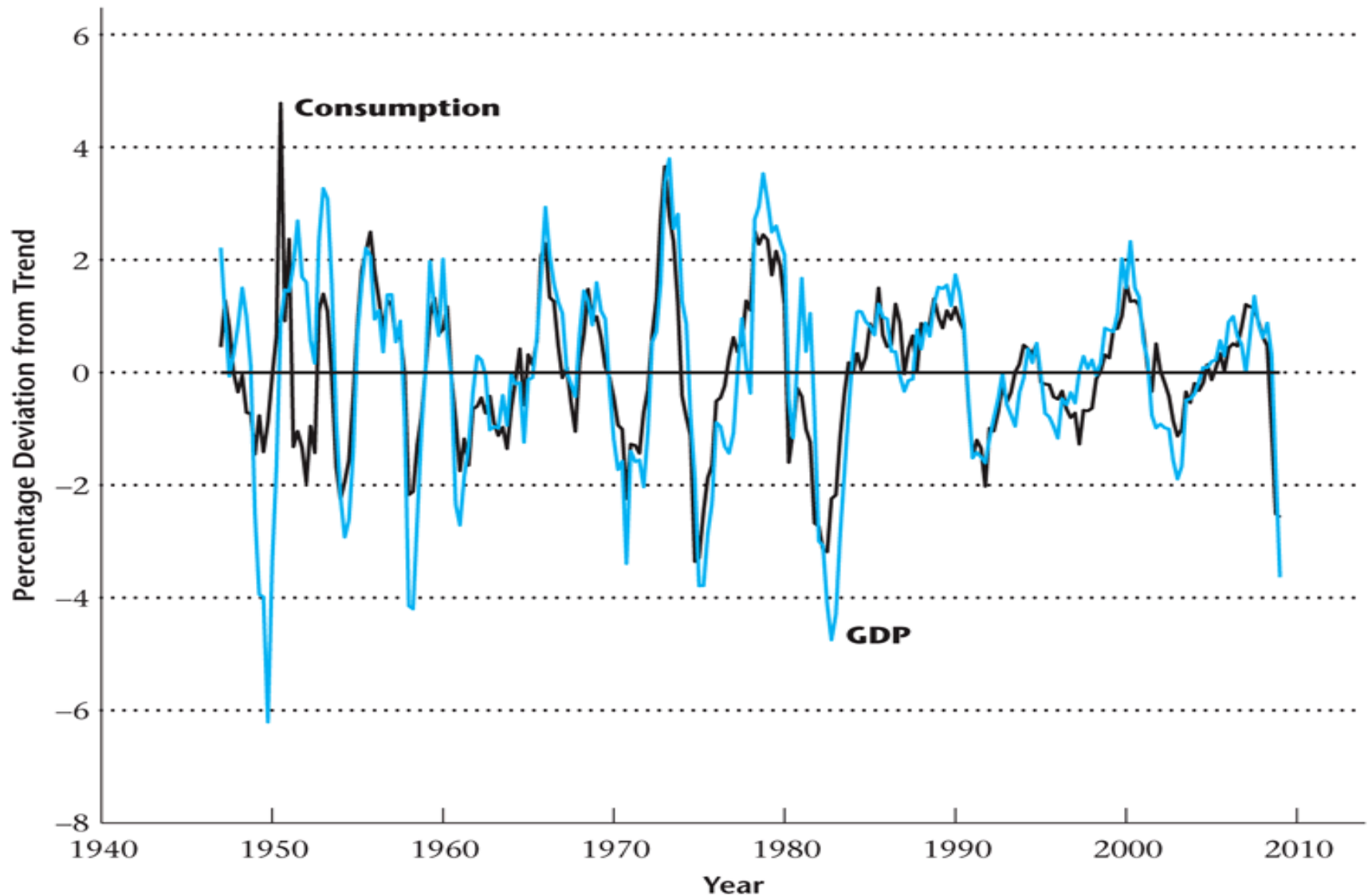
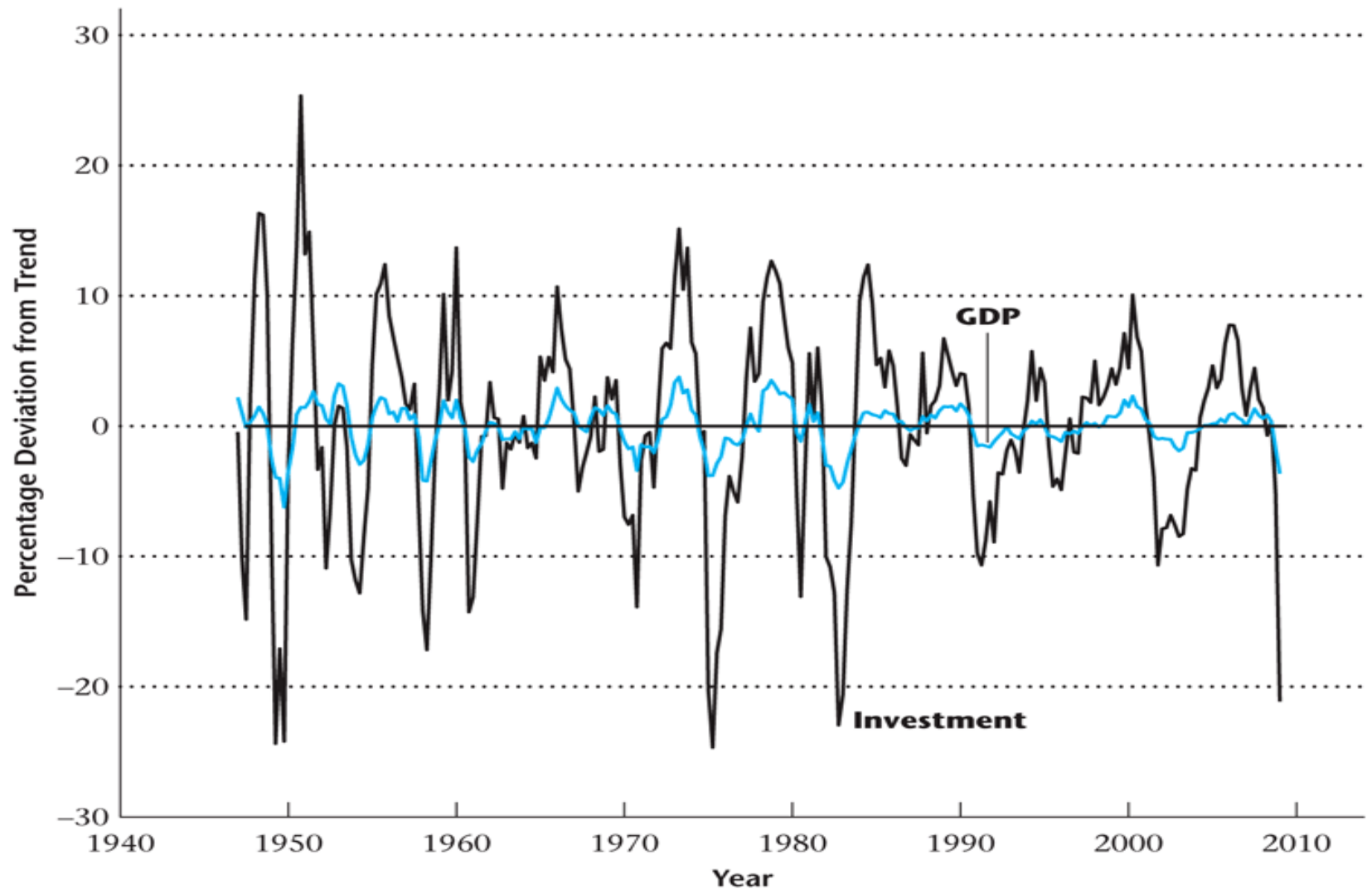


Figure 3.10: Percentage Deviations from Trend in Real Investment (black line) and Real GDP (colored line)



Nominal Variables and the Phillips Curve

- What is the correlation between money prices and aggregate economic activity?
- The Phillips curve is a negative relationship between the rate of change in money wages and the unemployment rate.

- If the unemployment rate to be a measure of aggregate economic activity, then the Phillips curve captures a positive relationship between the rate of change in a money price, the inflation rate, and the level of aggregate economic activity.
- The Phillips curve is not stable in data. We'll more of this issue in Chapter 18(5th edition) or in Chapter 15(6th edition).

Figure 3.11: Scatter Plot for the Percentage Deviations from Trend in the Price Level (the Implicit GDP Price Deflator) and Real GDP

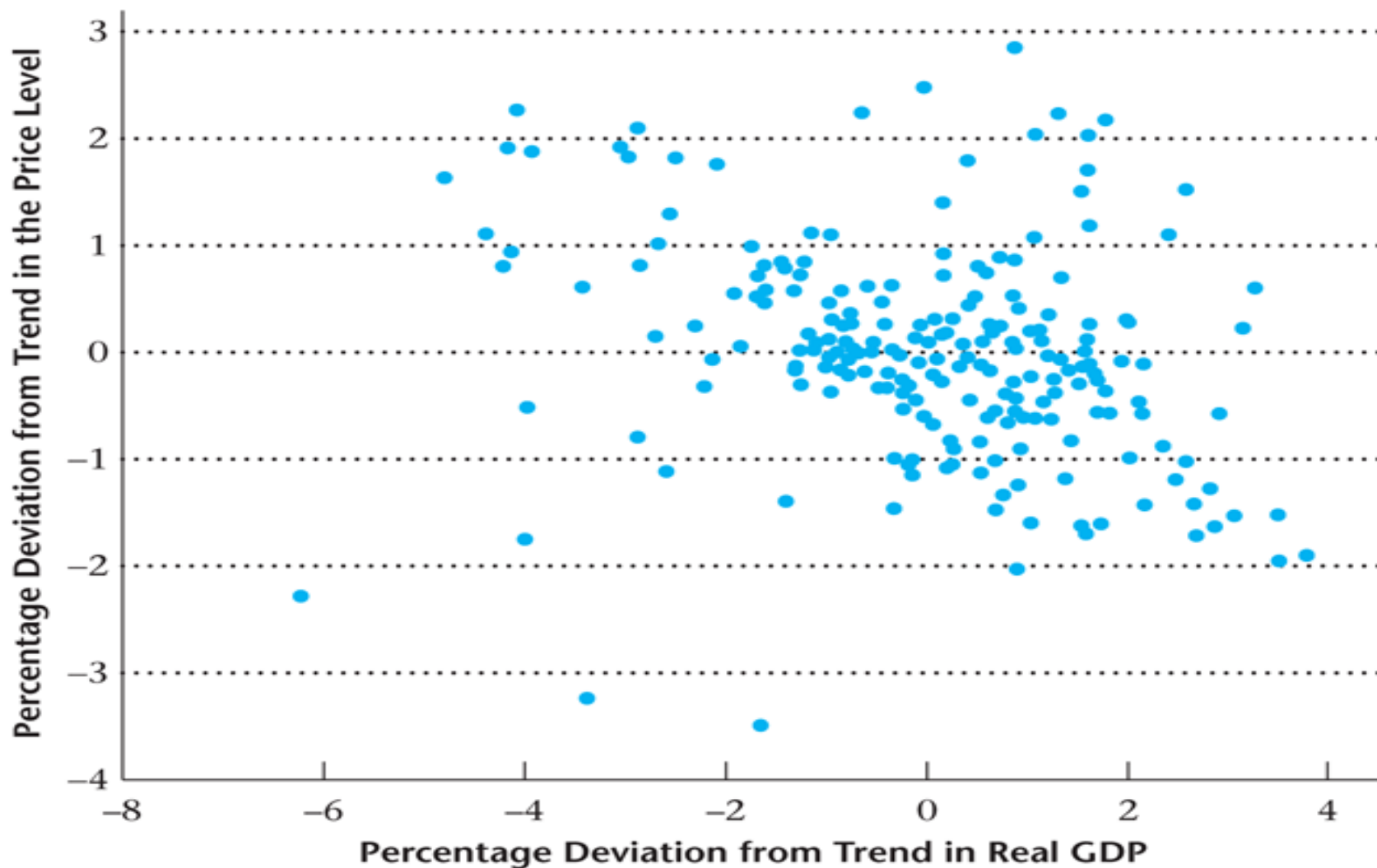


Figure 3.12: Price Level and GDP

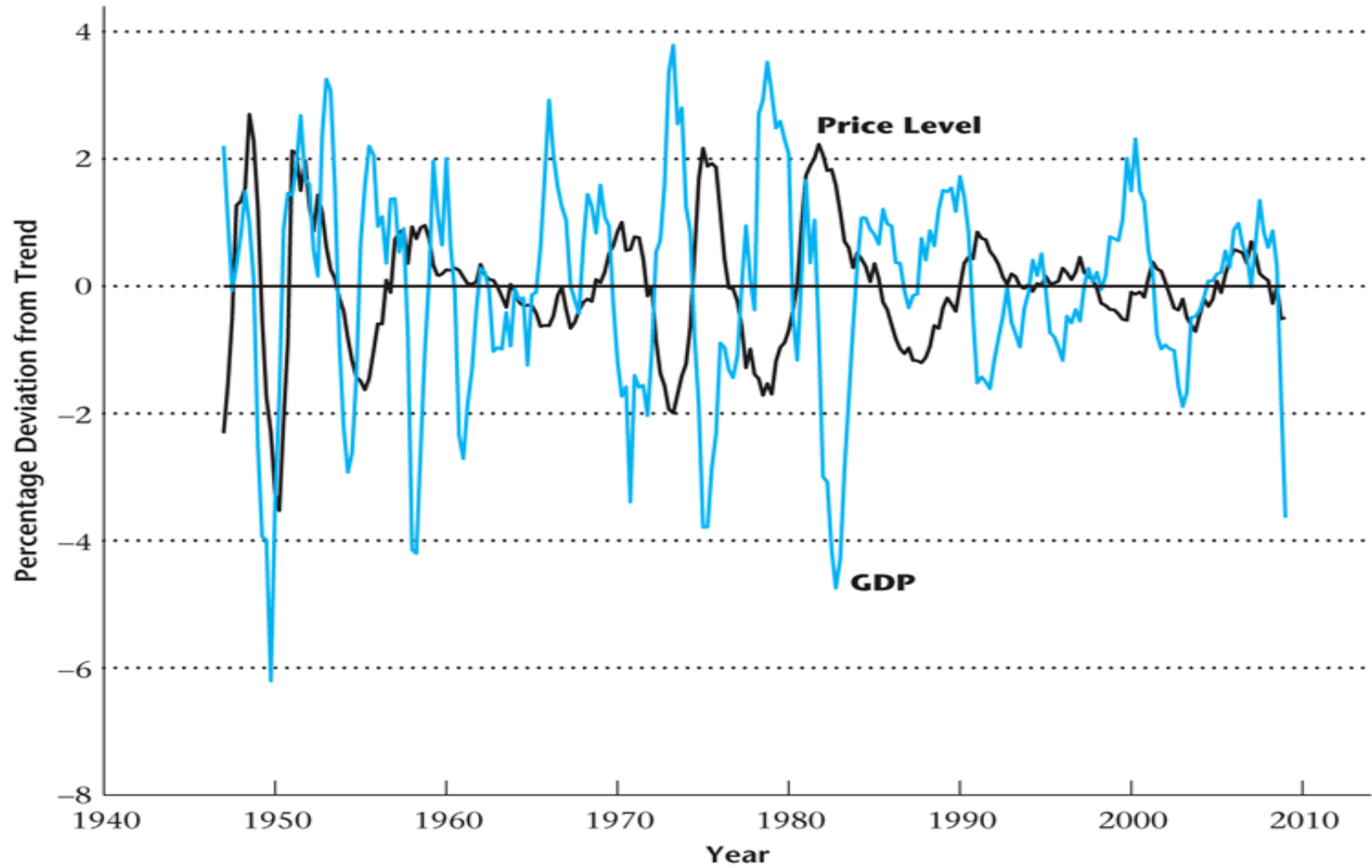


Figure 3.13: Percentage Deviations from Trend in the Money Supply (black line) and Real GDP (colored line) for the Period 1959–2009

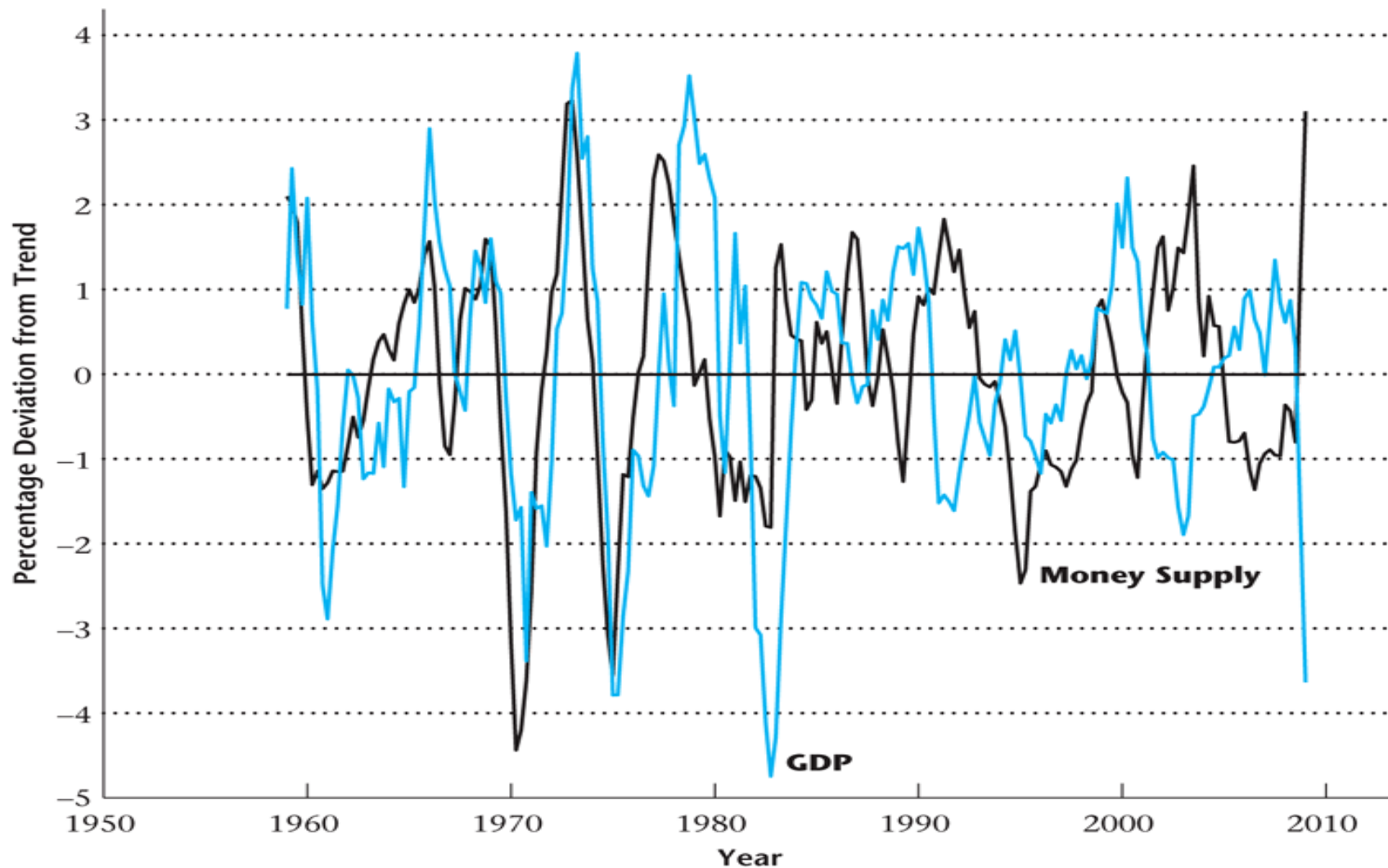


Figure 3.14: Percentage Deviations from Trend in Employment (black line) and Real GDP (colored line)

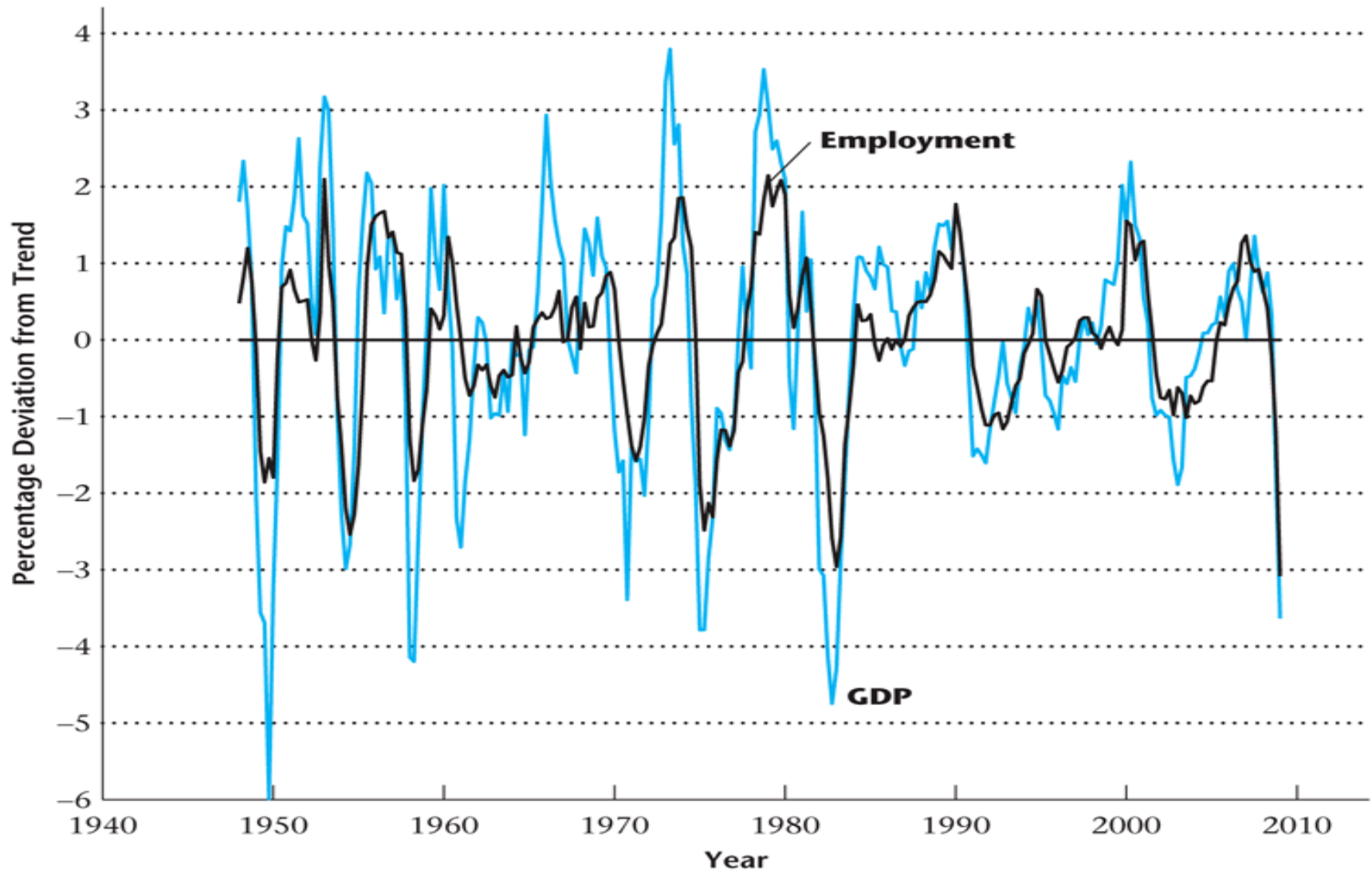


Figure 3.15: Percentage Deviations from Trend in Average Labor Productivity (black line) and Real GDP (colored line) for 1948–2009

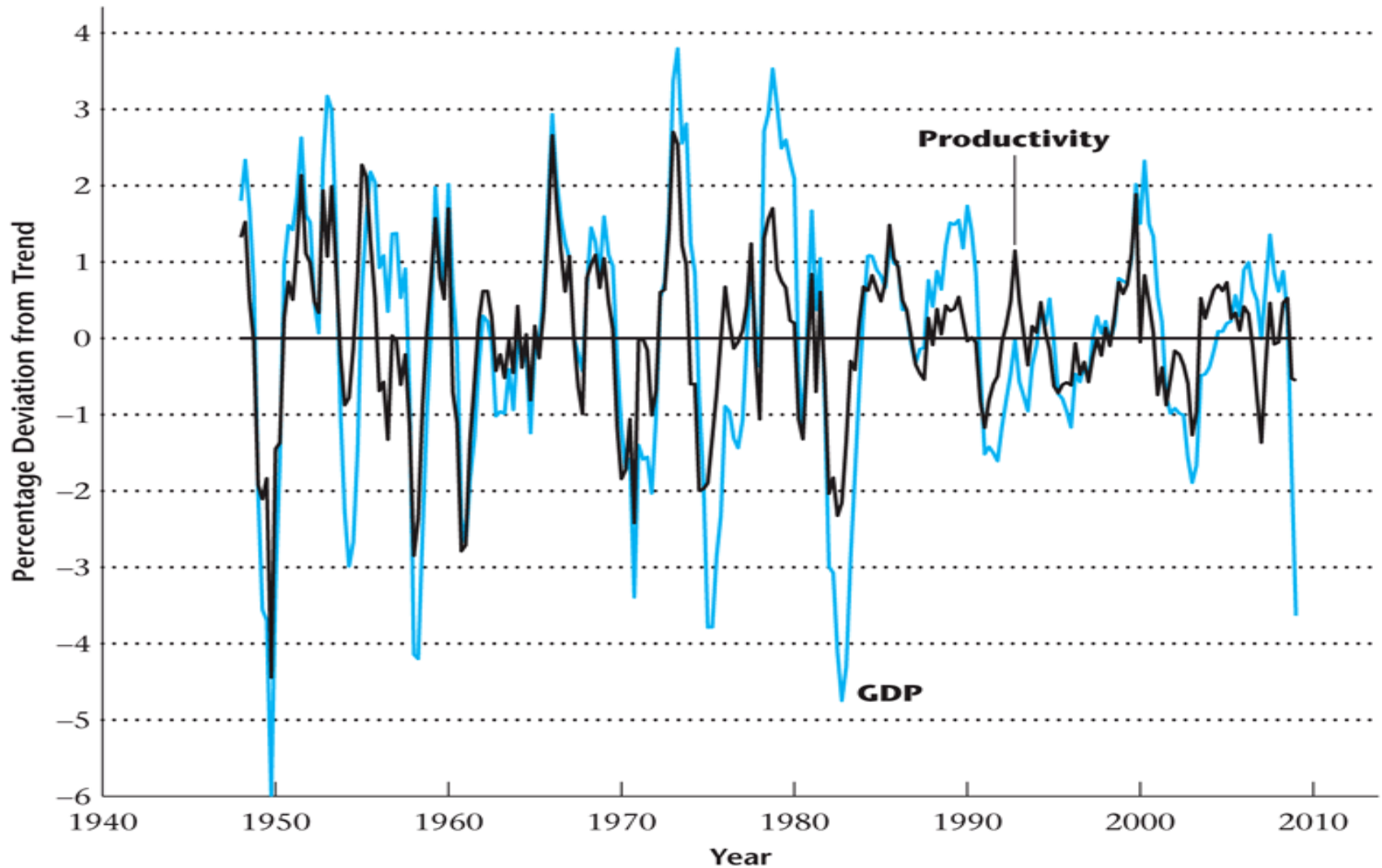


Figure 3.16: Seasonally Adjusted and Unadjusted Money Supply for the Years 2001–2009

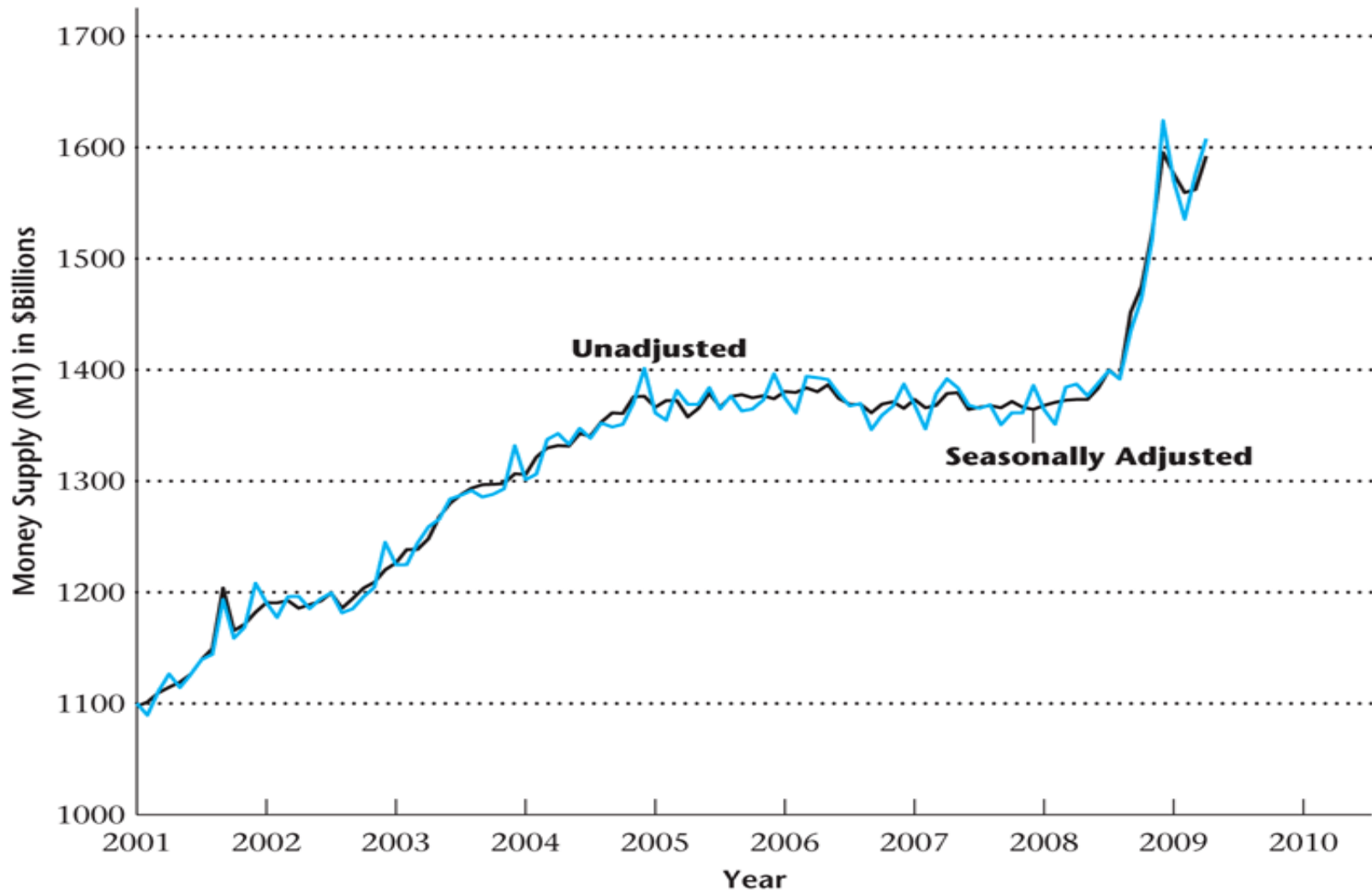


Table 3.1: Correlation Coefficients and Variability of Percentage Deviations from Trend

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	Correlation Coefficient	Standard Deviation (% of S.D. of GDP)
Consumption	0.76	75.9
Investment	0.84	478.9
Price Level	-0.23	57.4
Money Supply	0.26	80.4
Employment	0.80	61.5
Average Labor Productivity	0.81	62.4

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